

Propositional Logic Project

A group of **six travellers** is trapped inside the **Labyrinth of Truth and Lies**. To escape, they must correctly determine who among them is telling the truth and who is lying. The labyrinth contains magical doors that open only when specific logical conditions are met. However, there's a catch—**some travellers always tell the truth, while others always lie**.

Rules of the Labyrinth:

1. **Each traveller makes a statement, but you don't know who a truth-teller is and who is a liar.**
2. **A door will open only if a valid configuration of truth and lies satisfies the logical conditions.**
3. **There is at least one liar and one truth-teller in the group.**
4. **Your task is to determine the correct truth values of each traveller based on their statements and open the door.**

Statements of the Travelers:

- **Traveler A:** "Traveler B is lying."
- **Traveler B:** "Traveler C and D are both truth-tellers."
- **Traveler C:** "If A is truthful, then E is lying."
- **Traveler D:** "Traveler A and B are either both truthful or both lying."
- **Traveler E:** "At least two of us are lying."
- **Traveler F:** "Traveler D is truthful, and Traveler E is lying."

You have to submit the answers of the following:

1. Represent the Statements Using Propositional Logic
2. Construct the Truth Table
3. Implement a Python Solver
4. Find the Unique Escape Condition